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1 using System;
2
3 //                                Mostafa Abdelhay Ahmed Mansour
4 class Program
5 {
6     static int[] data = { 115, 182, 191, 31, 196, 1099, 5, 172, 10,
7                          179, 83, 21, 20, 21,
8                          186, 177, 195, 193, 188, 199, 62, 109, 105,
9                          183, 110 };
10
11 static void Main()
12 {
13     int n = data.Length;
14     double[] sorted = data.Select(x => (double)x).OrderBy(x => x).ToArray();
15
16     Console.WriteLine("\t\t\t\t\t==== Part 1: Statistics =====\n");
17     Console.WriteLine($"n = {n}");
18     Console.WriteLine($"Data (sorted): {string.Join(", ", sorted)}\n");
19
20     // (i) Mean
21     double mean = sorted.Average();
22     Console.WriteLine("(i)    Mean                = {mean:F4}");
23
24     // (ii) Mode
25     var mode = data.GroupBy(x => x)
26                  .OrderByDescending(g => g.Count())
27                  .First();
28     Console.WriteLine("(ii)   Mode                = {mode.Key}\n\n      (appears {mode.Count()} times)");
29
30     // (iii) Median
31     double median = Percentile(sorted, 50);
32     Console.WriteLine("(iii) Median              = {median:F4}");
33
34     // (iv) Variance (population variance)
35     double variance = sorted.Sum(x => Math.Pow(x - mean, 2)) / n;
36     Console.WriteLine("(iv)   Variance            = {variance:F4}");
37
38     // (v) P20
39     double p20 = Percentile(sorted, 20);
40     Console.WriteLine("(v)    P20                 = {p20:F4}");
41
42     // (vi) P50
43     double p50 = Percentile(sorted, 50);
44     Console.WriteLine("(vi)   P50                 = {p50:F4}");
45
46     // (vii) First Quartile Q1 = P25
47     double q1 = Percentile(sorted, 25);
48     Console.WriteLine("(vii) First Quartile Q1 = {q1:F4}");
49 }

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48 // (viii) Second Quartile Q2 = P50
49 double q2 = Percentile(sorted, 50);
50 Console.WriteLine($"(viii) Second Quartile Q2 = {q2:F4}");
51
52 // (ix) Third Quartile Q3 = P75
53 double q3 = Percentile(sorted, 75);
54 Console.WriteLine($"(ix) Third Quartile Q3 = {q3:F4}");
55
56 // (x) Range
57 double range = sorted[n - 1] - sorted[0];
58 Console.WriteLine($"(x) Range = {range:F4}");
59
60 // (xi) Interquartile Range
61 double iqr = q3 - q1;
62 Console.WriteLine($"(xi) Interquartile Range= {iqr:F4}");
63
64 // (xii) Standard Deviation
65 double stdDev = Math.Sqrt(variance);
66 Console.WriteLine($"(xii) Standard Deviation = {stdDev:F4}");
67
68 // (xiii) Summation of Divisions (sum of each value divided by
69 mean)
70 double sumDivisions = sorted.Sum(x => x / mean);
71 Console.WriteLine($"(xiii) Summation of Div. = {sumDivisions:F4}");
72
73 // Part 2: Outlier Detection using IQR method
74 Console.WriteLine("\n\t\t\t\t\t==== Part 2: Outlier Detection
75 (IQR Method) =====\n");
76 double lowerBound = q1 - 1.5 * iqr;
77 double upperBound = q3 + 1.5 * iqr;
78 Console.WriteLine($"Lower Bound = Q1 - 1.5*IQR = {lowerBound:F4}");
79 Console.WriteLine($"Upper Bound = Q3 + 1.5*IQR = {upperBound:F4}
80 \n");
81
82 foreach (int val in data)
83 {
84     bool isOutlier = val < lowerBound || val > upperBound;
85     Console.WriteLine($" {val,5} -> {(isOutlier ? "OUTLIER " :
86 "Normal")});");
87 }
88
89 // Interpolation-based percentile (same method used in statistics
90 textbooks)
91 static double Percentile(double[] sorted, double p)
92 {
93     int n = sorted.Length;
94     double L = (p / 100.0) * n;
95     int lower = (int)Math.Floor(L);
96     int upper = lower + 1;
97

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94         if (lower <= 0) return sorted[0];
95         if (upper >= n) return sorted[n - 1];
96
97         return sorted[lower - 1] + (L - lower) * (sorted[lower] - sorted ↵
           [lower - 1]);
98     }
99 }
```